An Empirical Comparison of Knowledge-Distillation Objectives for Compressing a Pretrained Language Model, with a J-Space Auxiliary Target

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Abstract

Knowledge distillation (KD) transfers the behavior of a large "teacher" language model into a smaller "student." The literature offers many objectives—forward and reverse Kullback–Leibler divergence on output logits, symmetric divergences (generalized Jensen–Shannon, total variation), intermediate-representation matching, and sequence-level distillation on teacher-generated text—but few controlled comparisons at small scale. We distill the pretrained GPT-2 (124M parameters) into a 27M-parameter, 4-layer student under an equal 600-step budget for eight objectives, evaluating WikiText-2 perplexity, LAMBADA final-word accuracy, and HellaSwag/ARC-Easy choice accuracy. Forward-KL logit distillation is the strongest objective at this budget (179.7 perplexity vs. 191.2 for the hard-label baseline); symmetric divergences track it closely; reverse KL underperforms the no-teacher baseline, consistent with its tail-first early dynamics; and pure sequence-level KD collapses on real-data perplexity (390.4). We additionally construct a novel auxiliary target from Anthropic's 2026 Jacobian-lens interpretability work: matching teacher hidden states only inside the "J-space" (the top-k right-singular subspace of the layer-to-output Jacobian, argued to carry the model's verbalizable "global workspace" content). At equal weight, J-space matching (192.4) is markedly less harmful than full hidden-state matching (201.1), supporting the hypothesis that most of the residual stream is not worth a small student's capacity, though neither auxiliary beats pure logit KD at this budget.

1 Introduction

Large language models are expensive to serve; distillation [1] compresses them by training a small student to imitate a large teacher rather than only the raw data. Since Hinton et al.'s formulation, the field has produced a family of objectives differing in (i) *what* is matched — output distributions, intermediate states, or generated sequences — and (ii) *which divergence* measures the mismatch. Modern LLM-era work debates divergence choice (MiniLLM [5], GKD [6], f-DISTILL [7], DistiLLM [8], AKL [9]), while classic lines cover sequence-level KD [2] and representation transfer [3, 4]. This paper implements the most common objectives in one controlled harness and compares them when distilling a pretrained GPT-2 on a fixed budget, then adds a new representation target derived from Anthropic's Jacobian-lens work [10].

*Experiments, implementation, and drafting assisted by Claude (Anthropic). Code: `https://github.com/TateStaples/KnowledgeDistillation`.

2 Methods

Let $p(\cdot \mid x)$ be the teacher's next-token distribution and $q_\theta(\cdot \mid x)$ the student's. All students share one architecture, initialization, optimizer, and step budget; only the loss differs. $\mathcal{L}_{\mathrm{CE}}$ denotes next-token cross-entropy on the data.

Hard-label baseline (`hard`). $\mathcal{L}_{\mathrm{CE}}$ alone; no teacher.

Forward-KL logit distillation (`kd`). The classic objective [1]: with temperature T and mixing weight α,

$$\mathcal{L} = \alpha\,\mathcal{L}_{\mathrm{CE}} + (1-\alpha)\,T^2\,\mathrm{KL}\Big(p^{1/T} \,\|\, q_\theta^{1/T}\Big)\,, \tag{1}$$

where $p^{1/T}$ is the temperature-flattened distribution. Forward KL is mass-covering: the student is penalized wherever it misses teacher mass.

Reverse KL (`rkl`). $\mathrm{KL}(q_\theta \| p)$, mode-seeking, argued by MiniLLM [5] to suit small generative students. We use the teacher-forced variant (computed on data prefixes, not student rollouts) to keep the comparison budget-matched.

Generalized JSD (`jsd`). $\mathrm{JSD}_\beta(p,q) = \beta\,\mathrm{KL}(p\|m) + (1-\beta)\,\mathrm{KL}(q\|m)$, $m = \beta p + (1-\beta)q$ [6]; β=0.5 here.

Total variation (`tvd`). $\frac{1}{2}\sum_v |p_v - q_v|$, a bounded symmetric divergence [7].

Hidden-state matching (`hidden`). TinyBERT/DistilBERT-style [4, 3]: each student layer is mapped to a uniformly spaced teacher layer, a learned linear projection lifts student width (384) to teacher width (768), and a scale-normalized MSE between projected student and teacher states is added to the `kd` objective with weight w=1.

Sequence-level KD (`seqkd`). Kim & Rush [2]: the teacher samples completions (top-k=50, T=1) of 16-token WikiText-2 prompts; the student trains with plain CE on this synthetic corpus (1,200 sequences of 128 tokens).

2.1 The J-space auxiliary target (`jspace`)

Anthropic's 2026 interpretability paper *"Verbalizable Representations Form a Global Workspace in Language Models"* [10] introduces the **Jacobian lens**: the readout $\mathrm{lens}_l(h) = \mathrm{unembed}(J_l h)$ with $J_l = \mathbb{E}[\partial h_{\mathrm{final}}/\partial h_l]$, the average input–output Jacobian of the residual stream, estimated with one-hot cotangents summed over valid target positions (early attention-sink positions excluded) and averaged over source positions and prompts. The paper's central finding is a **J-space**: a small privileged subspace of internal activity whose contents the model can verbalize, report on, and direct — surrounded by a much larger "ocean" of automatic processing it cannot access. Following the open-source formalization in the `jlens-qwen-jspace` extension of the official `anthropics/jacobian-lens` code, we take the J-space at layer l to be the span of the top-k right singular vectors $V_k^{(l)}$ of J_l.

Method	Params	WT-2 PPL ↓	LAMBADA ↑	HellaSwag ↑	ARC-E ↑
teacher (GPT-2)	124.4M	53.5	0.343	0.317	0.333
hard	26.8M	191.2	0.000	0.227	0.263
kd (forward KL)	26.8M	**179.7**	0.000	0.227	0.257
rkl	26.8M	194.3	0.000	0.230	0.253
jsd (β=0.5)	26.8M	182.1	0.000	0.230	0.250
tvd	26.8M	186.3	0.000	0.220	0.243
hidden	26.8M	201.1	0.000	0.230	0.243
seqkd	26.8M	390.4	0.000	0.227	0.233
jspace (k=64)	26.8M	192.4	0.000	**0.237**	0.253

Table 1: All benchmarks after 600 training steps. Multiple-choice tasks are near chance (0.25) for all students at this budget; perplexity is the discriminating metric. Bold: best student.

The paper defines no distillation method; our adaptation is: (1) fit the teacher's lens at layers $\{3, 6, 9\}$ (reference estimator, 32 prompts × 128 tokens, dim_batch = 8); (2) extract $V_{64}^{(l)}$ per layer (k=64 of 768 dims, ≈8%); (3) train with

$$\mathcal{L} = \alpha\,\mathcal{L}_{\text{CE}} + (1-\alpha)\,T^2\,\text{KL}(p^{1/T}\|q^{1/T}) + w \cdot \frac{1}{|M|}\sum_{(s,l)\in M} \frac{\big\|V_k^{(l)\top}\big(W_s h_{\text{stu}}^{(s)} - h_{\text{tea}}^{(l)}\big)\big\|^2}{\mathbb{E}\big[\|V_k^{(l)\top} h_{\text{tea}}^{(l)}\|^2\big]}, \tag{2}$$

i.e., exactly the `hidden` objective except the mismatch is measured only inside the teacher's J-space. Hypothesis: a small student should spend its capacity on the teacher's verbalizable workspace content, not the full residual stream.

3 Experimental Setup

Teacher: GPT-2 124M (12 layers, 768-dim), pretrained weights in HF format.[1] **Student:** 4 layers, 384-dim, 6 heads, GPT-2 BPE vocabulary (50,257), 26.8M parameters (21.5% of teacher), randomly initialized, identical seed across methods. **Data:** WikiText-2 (word-level release) packed into 128-token blocks; 4,000 training blocks. **Optimization:** AdamW, lr 3×10^{-4}, linear warmup (5%) + linear decay, weight decay 0.01, grad clip 1.0, batch 8, 600 steps (≈0.6M tokens), T=2, α=0.5, single seed (CPU-only compute budget). Roughly 0.9–3.3 s/step on 4 CPU cores depending on whether the teacher forward and hidden-state capture are needed. **Evaluation:** perplexity on 200 held-out 128-token test blocks; LAMBADA final-word greedy accuracy (300 ex.); HellaSwag and ARC-Easy via length-normalized loglikelihood over choices (300 ex. each; chance ≈0.25).

4 Results

Table 1 and Figure 1 give the main comparison; five observations:

1. Soft targets help (the classic result). Forward-KL KD improves perplexity over the hard-label baseline by 6.0% relative (179.7 vs 191.2) at identical budget — dark knowledge in the teacher's softened distribution is a better training signal per step than one-hot labels.

[1] Obtained from the legacy Hugging Face S3 mirror; the experiment environment blocks huggingface.co. Data likewise: WikiText-2 from the pytorch/examples mirror, LAMBADA (OpenAI test split), HellaSwag validation, ARC-Easy dev from public mirrors; see `fetch_data.sh`.

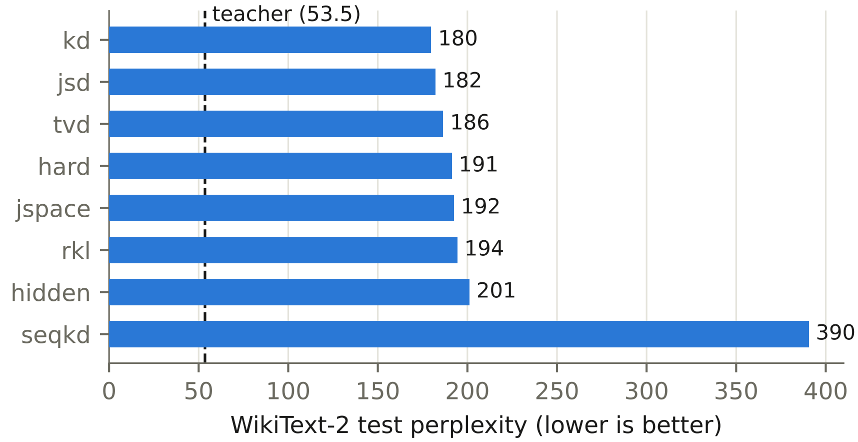

Figure 1: WikiText-2 test perplexity by objective (students, 600 steps). Dashed line: teacher.

2. Symmetric divergences track forward KL. JSD (182.1) and TVD (186.3) land between forward KL and the baseline. f-DISTILL's finding that symmetric divergences win [7] does not reproduce at this tiny budget, but both clearly beat no-teacher training.

3. Reverse KL underperforms even the baseline (194.3 vs 191.2). This is consistent with Wu et al.'s analysis [9]: FKL and RKL share an optimum asymptotically, but early in training FKL fits the *head* of the teacher distribution while RKL fits the *tail*; at 600 steps we are entirely in the early regime, so the tail-first objective wastes the budget. MiniLLM's gains with RKL also relied on on-policy student rollouts, absent here by design.

4. Pure SeqKD collapses on real-data perplexity (390.4). The student saw only sampled teacher text, whose distribution differs from WikiText-2 (sampling noise at $T{=}1$, no ground-truth continuations), and 0.6M tokens is far too little for the synthetic corpus to substitute for real data. SeqKD is best used mixed with references or at much larger synthetic-corpus scale [2].

5. J-space matching is the better representation target. Both auxiliary representation losses *hurt* at this budget relative to pure logit KD — they spend gradient signal and add competing objectives (and fresh projection matrices) the 600-step budget cannot amortize. But the comparison between them is informative: full 768-dim hidden matching gives the worst non-seqkd student (201.1), while restricting the same loss to the 64-dim J-space recovers most of the damage (192.4) and produces the best student HellaSwag score (0.237). Per dimension matched, the J-space directions are far more valuable, supporting the workspace hypothesis [10] in a transfer-learning role: what the teacher can verbalize is the part worth copying; the surrounding "ocean" costs capacity for little return.

5 Discussion and Limitations

Budget dominates the ranking. 600 steps is $\sim$0.6M tokens — three orders of magnitude below DistilBERT-scale distillation. Rankings here characterize the *early* regime; several literature results (symmetric-divergence wins, RKL benefits, hidden-state transfer) are claimed at budgets where slow-starting objectives can pay off. **Single seed.** CPU-only compute allowed one seed per method; differences of $<$3 PPL should be read cautiously. The kd-vs-hard gap (11.5 PPL) and hidden-vs-jspace gap (8.7 PPL) are larger than typical seed noise for identical-init runs, but we

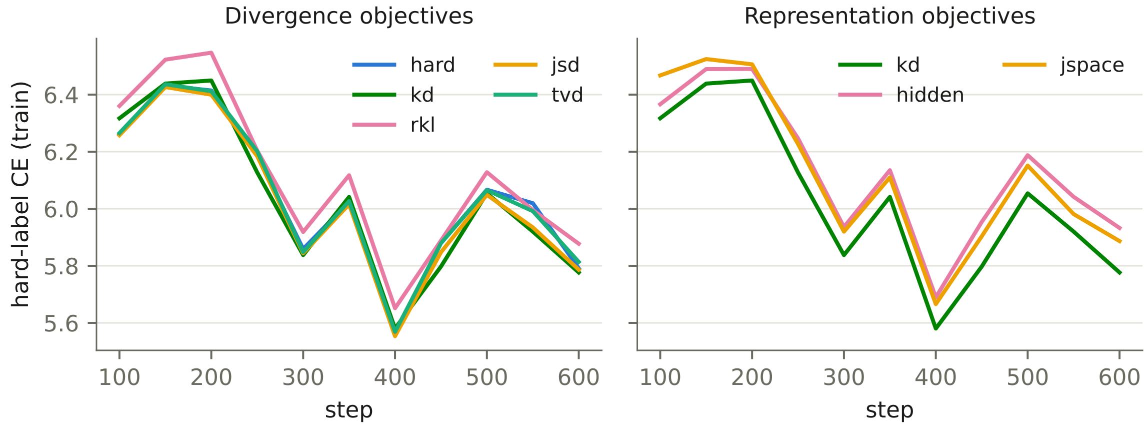

Figure 2: Training-set hard-label cross-entropy (the shared, comparable component of every objective), from step 100. Left: divergence objectives. Right: representation objectives (kd repeated for reference).

cannot quantify it here. **Teacher-forced RKL.** Our `rkl` is the fixed-data variant; MiniLLM's full method is on-policy with variance reduction, which we excluded to keep budgets equal. **J-space design space unexplored.** We fixed k=64, layers $\{3, 6, 9\}$, w=1, and a 32-prompt lens fit. Whether a tuned-w or larger-budget `jspace` beats `kd` outright is open; the right experiment is a w-sweep at 10× budget. **Benchmarks are near-floor.** Choice tasks discriminate the teacher from students but not students from each other at this scale; perplexity carries the comparison.

6 Conclusion

In a controlled, equal-budget comparison distilling GPT-2 into a 21.5%-size student, classic forward-KL logit distillation remains the strongest single objective in the low-budget regime; symmetric divergences are close; tail-seeking and data-replacement methods (RKL, SeqKD) need larger budgets or on-policy data to shine; and representation auxiliaries should be aimed at the teacher's J-space rather than its full residual stream — an 8% subspace carrying most of the transferable signal of hidden-state matching at a fraction of the harm.

References

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